PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Protocol for updated systematic review and meta-analysis on the burden of non-communicable diseases among people living with HIV in Sub-Saharan Africa
AUTHORS	Moyo, Maureen; Musekiwa, Alfred

VERSION 1 – REVIEW

REVIEWER	Todowede, Olamide University of York, Department of Health Science
REVIEW RETURNED	07-Sep-2021

GENERAL COMMENTS	The comorbidity of NCD and HIV is an interesting topic and requires more information. Especially in SSA countries that are experiencing double burden of infectious diseases and NCD. The authors should consider the following; 1. I will suggest you compare communicable diseases and NCD, rather than compare HIV/AIDS and NCDs or discuss how NCD are common among HOV population. 2. 'Low- and middle-income countries' not low- and medium-income countries 3. Line 61 …infectious diseases not multiple …….. 4. The title and abstracts of the identified observational studies published since 2010 will be screened for mention of HIV and any NCD/NCD risk. – These needs rewording. 5. Outcomes of interest are prevalence or incidence of any NCD/NCD risk factors – I will suggest you add in HIV population 6. Line 62 – 66, could the author reword this line. Some information in there is not true and need to be supported with references. NCD has nothing to do with universal testing and treatment for HIV. 7. The justification for wanting to update the systematic review is quite weak, the author should research more what has changed between 2018 and 2021, which calls for an update of the review. It could be that the trends are increasing or reducing. 8. Search Strategy- The focus of the study is Sub-Sahara Africa, I do not think you need to add LMIC in the search string, SSA and the list of SSA countries should suffice. 9. Line 254 …The author needs to clarify which mental illness is of interest to the study, it will be important to check the common comorbid mental disorder among PLHIV for this review. 10. The Author should consider doing an analysis that will compare the systematic review that they are updating and the present result to see if there are differences in the cumulative prevalence/incidence of NCD among PLHIV post-Patel 2018. 11. In the Data Synthesis section, the author did not state how they plan to analyse the association of ART duration and ART use.

REVIEWER	Mutanga, Jane N US Food and Drug Administration
REVIEW RETURNED	07-Sep-2021

GENERAL COMMENTS	This is a timely review protocol. The flow is clear to me. I have made a few suggestions for your consideration. Abstract Line 21: instead of determine, consider using the word estimate or another synonym. Line: 26-28: please be clear with your exposure of interest, are you only interested in PLHIV who are taking ART, HIV is not of interest to you. I believe that from 2010, there are studies of PLHIV not on ART? Be clear on this. Introduction: Line 61: Can you say how COVID has impacted NCDs and HIV in SA? Line 62: Are there any social economic milestones which can also be attributed to increase in NCDs in the general population? Line 69: You might want to investigate your classification of risk factors. From a medical point of view, I am familiar with modifiable and non-modifiable risk factors, and metabolic syndrome. This classification from the WHO is confusing. Line 95: Contracting NCDs: reconsider use of the word contract, this implies that NCDs are passed from person to person like an infectious disease. Line 98: Can't start a sentence with an abbreviation Aims: Line 115: Objective 4 is not clear. This is a loaded statement, please try to unpack it. Are you trying to assess if PLHIV who have been on ART for a longer time are more likely to have NCDs? Then you have to compare with controls who are not living with HIV and look at confounders and effect modifiers. Line 117: How is objective 3 different from objective 4? Methods: Line 127: Are you excluding PLHIV not on ART? Line 128: will RCTs be excluded? Line 133: Your search appears very limited; is there any reason you are excluding the other databases like web of science etc.?

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Olamide Todowede, University of York
Comments to the Author:
The comorbidity of NCD and HIV is an interesting topic and requires more information. Especially in SSA countries that are experiencing double burden of infectious diseases and NCD. The authors should consider the following;

1. I will suggest you compare communicable diseases and NCD, rather than compare HIV/AIDS and NCDs or discuss how NCD are common among HOV population.
We will discuss how NCDs are common among PLHIV as this was the focus of the research, which is part of a PhD on integration of NCD services with HIV services in primary healthcare facilities of Southern African countries.

2. 'Low- and middle-income countries' not low- and medium-income countries
Line 18: "Low- and medium- income countries" is now replaced with "Low- and middle-income

countries'"

3. Line 61 …infectious diseases not multiple …….
Line 60: "multiple" has now been deleted

4. The title and abstracts of the identified observational studies published since 2010 will be screened for mention of HIV and any NCD/NCD risk. – These needs rewording.
The sentence has now been reworded
Line 24-25 & 140 – 141 : "Two authors will independently screen the title and abstracts of the articles identified from the search."

5. Outcomes of interest are prevalence or incidence of any NCD/NCD risk factors – I will suggest you add in HIV population
"in HIV population" has now been added
Line 142-143 : "Outcomes of interest are prevalence or incidence of any NCD or NCD risk factors in HIV populations."

6. Line 62 – 66, could the author reword this line. Some information in there is not true and need to be supported with references. NCD has nothing to do with universal testing and treatment for HIV.
The lines have now been reworded.
Line 77-84: The increase in NCDs has been attributed to physical inactivity, unhealthy diets, harmful use of alcohol and tobacco in LMICs, just as in the developed countries 3. Lifestyle changes in SSA due to the growth of urbanisation, that entails more sedentary life style as urban work is often less physical, exposure to unhealthy diets that are high in salt, fat and sugar and pollution, among other factors have contributed to the NCD burden13. Among PLHIV, there have been reports that HIV itself or ART side-effects predisposing individuals further to NCDs14. Furthermore, the general increase in lifespan implies increased risk to age-related NCDs.

7. The justification for wanting to update the systematic review is quite weak, the author should research more what has changed between 2018 and 2021, which calls for an update of the review. It could be that the trends are increasing or reducing.
Line 117 – 123 : "Patel et al (2018)2, a systematic review, included articles published between 2010 and 2016 on the burden of NCDs among PLHIV in LMICs. However, it has not been updated and there have been several publications between 2017 and 2021. This systematic review will update the evidence from 2010 to the present date and report on changes in trends of NCD burden in the HIV population. The updated systematic review will include chronic respiratory diseases that were excluded in the previous systematic review2."

8. Search Strategy- The focus of the study is Sub-Sahara Africa, I do not think you need to add LMIC in the search string, SSA and the list of SSA countries should suffice.
Page 9 : This is noted. The "LMIC" has now been removed from the search string.

9. Line 254 …The author needs to clarify which mental illness is of interest to the study, it will be important to check the common comorbid mental disorder among PLHIV for this review.
We will consider depression as the common comorbid mental disorder among PLHIV for this review.
Line 263-269: A pre-designed data extraction form will be used by two review authors who will independently extract data on prevalence or incidence of any NCD (cardiovascular disease, cancer, mental illness(specifically , depression), chronic respiratory disease and diabetes) among PLHIV, prevalence or incidence of NCD risk factors (hypertension, hyperlipidaemia, dyslipidaemia, physical inactivity, obesity, smoking and pollution) in PLHIV, study design, sample size, participants' age, recruitment methods, study country and date of study publication.

10. The Author should consider doing an analysis that will compare the systematic review that they are updating and the present result to see if there are differences in the cumulative prevalence/incidence of NCD among PLHIV post-Patel 2018.
As part of the analysis we will compare the systematic review results with Patel 2018 as suggested.
Line 291-293: As part of the analysis we will compare the systematic review results with Patel 2018 and note any differences in the cumulative prevalence/incidence of NCDs among PLHIV post 2016.

11. In the Data Synthesis section, the author did not state how they plan to analyse the association of ART duration and ART use.
We have now decided to exclude the analysis of duration on ART as a risk factor for NCDs and/or NCD risk factors among PLHIV in SSA. We believe this should be analysed separately in a separate manuscript.
Page 7 : We have now removed objective 4.

Reviewer: 2
Dr. Jane N Mutanga, US Food and Drug Administration
Comments to the Author:
Dear Author,

This is a timely review protocol. The flow is clear to me. I have made a few suggestions for your consideration.

Abstract
Line 21: instead of determine, consider using the word estimate or another synonym.
“Determine” has now been changed to “estimate”.
Line 21: “This systematic review and meta-analysis will estimate the overall prevalence or incidence of NCDs (or its risk factors) among adults living with in HIV in SSA.”

Line: 26-28: please be clear with your exposure of interest, are you only interested in PLHIV who are taking ART, HIV is not of interest to you. I believe that from 2010, there are studies of PLHIV not on ART? Be clear on this.
We are interested in reporting the burden of NCDs among PLHIV irrespective of their ART status, meaning that we will consider studies with either PLHIV on ART or not on ART.
Line 26 – 28: Exposure of interest will be HIV (with or without ART). Outcomes of interest are prevalence or incidence of any NCD/NCD risk factors.

Introduction:
Line 61: Can you say how COVID has impacted NCDs and HIV in SA?
Line 71-76: The WHO has reported disruptions in NCD services due to the COVID-19 pandemic globally, including 41 African countries. The main reasons for the disruption to the NCD services are the decrease in inpatient volume due to cancellation of elective services, closure of population level screening, Government or public transport lock-downs hindering access to the health facilities for patients, NCD related clinical staff deployed to provide COVID-19 relief, closure of outpatient disease specific consultation clinics, insufficient personal protective equipment available for health care providers to provide services, insufficient staff to provide services, closure of outpatient NCD services as per government directive, decrease in outpatient volume due to patients not presenting, inpatient services/hospital beds not available and stock out of essential medicines, medical diagnostics or other health products at health facilities. This disruption in NCD services, has highlighted the need for LMICs such as SSA to strengthen their health systems, as a means of improving preparedness for future epidemics 12.

Line 62: Are there any social economic milestones which can also be attributed to increase in NCDs in the general population?
78-81: Lifestyle changes in SSA due to the growth of urbanisation, that entails more sedentary life style as urban work is often less physical, exposure to unhealthy diets that are high in salt, fat and sugar and pollution, among other factors have contributed to the NCD burden13.

Line 69: You might want to investigate your classification of risk factors. From a medical point of view, I am familiar with modifiable and non-modifiable risk factors, and metabolic syndrome. This classification from the WHO is confusing.
Line 85-88: Modifiable behavioural risk factors for NCDs include, tobacco use (including second-hand smoke), physical inactivity, unhealthy diet, and the harmful use of alcohol which consequently lead to raised blood pressure, overweight/obesity, hyperglycemia and hyperlipidaemia3. Environmental air pollution has also been identified as a key risk factor for NCDs in general 7.

Line 95: Contracting NCDs: reconsider use of the word contract, this implies that NCDs are passed from person to person like an infectious disease.
We have now replaced the word “contract” with “having”
Line 105 – 107:” It has previously been reported that PLHIV are at risk of having NCDs from HIV infection itself, from ART and from the risk associated with increasing age.”

Line 98: Can’t start a sentence with an abbreviation
The sentence has now been rewritten, removing the abbreviation from the start of the sentence.
Line 98-104: The NCDs are a challenge to economic development 3 7 14. Globally, 37 % of the deaths occur in people aged 30-69 years old, which is the working age group. As a result, employers face high staff turnover due to prolonged absenteeism and eventually deaths. Additionally, household costs associated with health care, increase in families that are already faced with financial difficulties in LMICs due to the chronic nature of the disease3 14. Indeed, NCDs tend to foster poverty as usually it is breadwinners whose lives are lost, thus leaving families in financial difficulties, especially in LMICs.

Aims:
Line 115: Objective 4 is not clear. This is a loaded statement, please try to unpack it. Are you trying to assess if PLHIV who have been on ART for a longer time are more likely to have NCDs? Then you have to compare with controls who are not living with HIV and look at confounders and effect modifiers.
We have now excluded objective 4
We are no longer considering the duration on ART associated with NCD/NCD risk factors among PLHIV. We believe this should be analysed separately in a separate manuscript.

Line 117: How is objective 3 different from objective 4?
Methods:
We have now excluded objective 4.

Line 127: Are you excluding PLHIV not on ART?
We will include all individuals living with HIV regardless of ART status.

Line 128: will RCTs be excluded?
Yes, RCTs will be excluded to minimise on bias due to the selection criteria of participants participating in RCTs.

Line 133: Your search appears very limited; is there any reason you are excluding the other

databases like web of science etc.?
We will now conduct searches on PubMed, EBSCOhost and Scopus .

VERSION 2 – REVIEW

REVIEWER	Mutanga, Jane N US Food and Drug Administration
REVIEW RETURNED	26-Feb-2022

GENERAL COMMENTS	I think this protocol is ready for publication. I did not spot any further problematic issues.